Ericsson PE Node ELEC-PE-06 Optical Transport Failure SOP (ALARM_9999)

Node: ELEC-PE-06 | **Vendor:** Ericsson | **Severity:** Warning | **Alarm Code:** ALARM_9999

Description

This Standard Operating Procedure (SOP) outlines the troubleshooting and resolution steps for Optical Transport Failures on the Ericsson PE Node ELEC-PE-06 at the Electronic City Hub. Optical Transport Failures in this context are primarily caused by degradation of SFP/XFP modules, fiber cuts, or loss of Rx/Tx power. The PE Node, being a critical component in the Optical Transport Network, relies heavily on the Synchronous Packet Transport (SPT) protocol to manage and transport packets across the network. SPT, running on top of the Optical Transport Network (OTN), enables the PE Node to efficiently manage and route packets between different network entities. The OTN, in turn, relies on the Optical Channel Transport Unit (OTU) to define the frame structure and transmission parameters for optical signals. Any degradation in the OTU or SFP/XFP modules can lead to Optical Transport Failures, resulting in ALARM_9999 being raised on the PE Node.

Pre-requisite Checks

- Ensure remote hands are dispatched to the Electronic City Hub to perform physical checks on the PE Node and associated equipment.

- Verify the status of redundant links and ensure they are functioning correctly.

- Check the power supply to the PE Node and ensure it is within the recommended operating range.

Potential Root Causes

Cause	Probability
Degradation of SFP/XFP modules	High
Fiber cuts or breaks	Medium
Loss of Rx/Tx power	Low

Diagnostic Commands & Expected Outputs

```
Command: show otu status, Expected Output: OTU Status: Normal,
Description: This command checks the status of the OTU and verifies if
it is functioning correctly. The expected output indicates a normal
OTU status.

Command: show sfp status, Expected Output: SFP Status: Normal,
Description: This command checks the status of the SFP/XFP modules and
verifies if they are functioning correctly. The expected output
indicates a normal SFP status.

Command: show power status, Expected Output: Power Status: Normal,
Description: This command checks the power supply to the PE Node and
```

```
verifies if it is within the recommended operating range. The expected
output indicates a normal power status.
```

Resolution Steps

Step 1: Step: Replace SFP/XFP module, Command: replace sfp module 1/1/1, Description: Replace the degraded SFP/XFP module with a new one. Follow the manufacturer's instructions for replacing the module.

Step 2: Step: Repair fiber cut, Command: repair fiber cut, Description: Repair the fiber cut or break. Follow the manufacturer's instructions for repairing the fiber.

Step 3: Step: Adjust Rx/Tx power, Command: adjust power 1/1/1, Description: Adjust the Rx/Tx power to the recommended operating range. Follow the manufacturer's instructions for adjusting the power.

Post-Check Verification

```
Command: show otu status, Expected Output: OTU Status: Normal,
Description: Verify that the OTU status is normal after the resolution
steps.

Command: show sfp status, Expected Output: SFP Status: Normal,
Description: Verify that the SFP status is normal after the resolution
steps.

Command: show power status, Expected Output: Power Status: Normal,
Description: Verify that the power status is normal after the
resolution steps.
```

Roll-back Procedures

- Step: Restore original SFP/XFP module, Command: restore sfp module 1/1/1, Description: Restore the original SFP/XFP module if the replacement module fails to resolve the issue.

- Step: Restore original fiber, Command: restore fiber cut, Description: Restore the original fiber if the repair fails to resolve the issue.

- Step: Restore original power settings, Command: restore power 1/1/1, Description: Restore the original power settings if the adjustment fails to resolve the issue.